EE605 - Error Correcting Codes Aug. 31st, 2023

Lecture 8

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In this lecture, we will focus on bounds on error correcting codes. By a bound, we mean a limit on some sort of performance of a class of codes. We will consider (n, M, d) codes in general and will specify if some particular bound is only for linear codes. Before beginning, let us review some definitions.

1 Preliminary Definitions

Definition 1.1 (Maximal Size)**.** Fix $n, d \in \mathbb{N}$. Define the maximal size of a $(n, \cdot, d)$ code over $\mathbb{F}_q$ as,

$$A_q(n, d) := \max\{|\mathcal{C}| : \quad \mathcal{C} \text{ is a } (n, M, d) \text{ code}\} \tag{1}$$

$\diamondsuit$

We will deal with bounds concerning the maximal code size. Also, another thing is relevant:

Definition 1.2 (Balls)**.** Let $c \in \mathbb{F}_q^n$, $r \in \mathbb{N}$ and define,

$$\mathcal{B}_r(c) := \{x \in \mathbb{F}_q^n : \ d(x, c) \leq r\} \tag{2}$$

$\diamondsuit$

This is the Hamming ball of size r centered around some vector c. With this, we define:

Definition 1.3 (Volume of a ball)**.** Consider the all-zeros vector $0 \in \mathbb{F}_q^n$ and define,

$$\text{Vol}_q(n, t) := |\mathcal{B}_t(0)| \tag{3}$$

$\diamondsuit$

Note that the volume of a ball of some fixed radius around any vector is the same, so we choose the all-zeroes vector in the definition for simplicity.

2 Bounds on Error Correcting Codes

We will first state and prove the singleton bound.

Theorem 1 (Singleton bound)**.** Consider a (n, M, d) code over the alphabet $\mathbb{F}_q$. We have that,

$$\boxed{d \leq n - \lceil \log_q M \rceil + 1} \tag{4}$$

Equivalently, $M \leq q^{n-d+1}$.

Note that for a (n, k, d) linear code, this reduces to $k \leq n - d + 1$, which implies $d \leq n - k + 1$. We now give two proofs, one for general codes and one for linear codes.

Proof. Define $l := \lceil \log_q M \rceil - 1$. Then, we have that

$$q^l = q^{\lceil \log_q M \rceil - 1} \leq q^{\log_q M + 1 - 1} = M \tag{5}$$

Note that q^l is the size of the set of all the l length vectors over $\mathbb{F}_q$. Since the set of codewords has more elements than q^l, this implies that there exist two codewords that agree in the first l places. Thus,

$$d \leq n - l = n - \lceil \log_q M \rceil + 1 \tag{6}$$

□

Now, let us prove the same bound, assuming a linear structure of the code.

Proof. Recall that the parity check matrix of a (n, k, d) linear code is of size $(n-k) \times n$. Thus, $\text{Rank}(H) \leq n - k$. Also, recall that any collection of $d - 1$ columns of H is linearly independent. Then we must have $d - 1 \leq n - k \implies k \leq n - d + 1$.
Aliter: Consider the generator matrix in systematic form $G = [I_n \mid A]_{k \times n}$. Note that the maximal weight of any row of this matrix is $1 + n - k$. By linearity and the definition of minimum distance, $d \leq \text{wt(row)} \leq n - k + 1$. □

Codes that achieve the singleton bound are called *Maximal Distance Separable* (MDS) codes. Some examples include,

1. The $(n, 1, n)$ repetition code.
2. The $(n, n+1, 2)$ simple parity check code.
3. **Reed-Solomon codes** over $\mathbb{F}_q$ with $q > n$. These are constructed as follows. Choose n distinct elements $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ (this is possible, why?), and construct the p.c. matrix as,

$$H := \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \alpha_1 & \alpha_2 & \cdots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \cdots & \alpha_n^2 \\ \vdots & \vdots & \vdots & \vdots \\ \alpha_1^{n-k-1} & \alpha_2^{n-k-1} & \cdots & \alpha_n^{n-k-1} \end{pmatrix}_{(n-k) \times n} \tag{7}$$

   It is easy to prove that $d = n-k+1$ here: We need to show that any $n-k$ columns are linearly independent, and there exists a set of $n-k+1$ linearly dependent columns. Note that this is a Vandermonde matrix by construction and thus has full rank,

$$\text{Rank}(H) = \min\{n-k, n\} = n - k \tag{8}$$

   Thus, any $n - k$ columns are linearly independent, and adding another column will cause linear dependence as the latter will be a linear combination of the former columns owing to the fact that the dimension of the column space is $n - k$.

MDS codes are popular in large storage systems. Moreover, they can handle $n-k$ erasures. That is, if one can just collect k correct received symbols, one can recreate the message sent!
Let us now discuss the famous Gilbert-Varshomov (GV) bound. We will outline two formulations: one is a bound proved similar to the bounds above, and one is more constructuve.

Theorem 2 (GV bound I)**.** For a $(n, \cdot, d)$ code over $\mathbb{F}_q$ we have,

$$A_q(n,d) \geq \frac{q^n}{\mathrm{Vol}_q(n,d-1)} \tag{9}$$

Proof. Consider the maximal (n, M, d) code $\mathcal{C}_0$ over $\mathbb{F}_q$, with $M := |\mathcal{C}_0| = A_q(n,d)$. Now, let $x \in \mathbb{F}_q^n$ be arbitrary. We claim that there exists $c_x \in \mathcal{C}_0$ s.t. $d(x,c) \leq d-1$. If not, then one could construct a code $\tilde{\mathcal{C}} := \mathcal{C}_0 \cup \{x\}$ which would be larger than $\mathcal{C}_0$, thus violating the maximal assumption. Thus, we have,

$$x \in \mathbb{F}_q^n \implies x \in \cup_{c \in \mathcal{C}_0} \mathcal{B}_{d-1}(c) \tag{10}$$

Thus, we have $q^n \leq | \cup_{c \in \mathcal{C}_0} \mathcal{B}_{d-1}(c)|$. Now,

$$| \cup_{c \in \mathcal{C}_0} \mathcal{B}_{d-1}(c) | \leq \sum_{c \in \mathcal{C}_0} |\mathcal{B}_{d-1}(c)| = A_q(n,d) \times \mathrm{Vol}_q(n,d-1) \tag{11}$$

by the union bound. Chaining these two inequalities we get,

$$A_q(n,d) \times \mathrm{Vol}_q(n,d-1) \geq q^n \tag{12}$$

giving

$$A_q(n,d) \geq \frac{q^n}{\mathrm{Vol}_q(n,d-1)}. \tag{13}$$

□

This give us a *lower* bound on the maximal code size, which is essentially possible due to a sphere covering argument. Recall that the Hamming bound gave us an upper bound using a sphere packing argument. Now we move on to the constructive version.

Theorem 3 (GV bound II)**.** If there exist $n, k, d, q \in \mathbb{N}$ satisfying

$$q^k < \frac{q^n}{\mathrm{Vol}_q(n-1,d-2)} \tag{14}$$

then there exists a (n,k,d) linear code.

Proof. We attempt to construct a $(n-k) \times n$ p.c. matrix for the (n,k,d) linear code. Start by the $I_{n-k \times n-k}$ identity matrix, and keep adding columns h_i. Suppose we have added the columns $h_1, h_2, \ldots, h_{l-1}$ and we need to add $h_l \in \mathbb{F}_q^{n-k}$ *such that* h_l is *not* a linear combination of any of $d-2$ columns from $h_1, \ldots, h_{l-1}$. Phrased differently, this condition reads,

$$h_l \notin \{[h_1, h_2, \ldots, h_{l-1}]x : x \in \mathbb{F}_q^{l-1},\ \mathrm{wt}(x) \leq d-2\} \tag{15}$$

The number of such columns, which is the number of ineligible vectors, is less than or equal to $\mathrm{Vol}_q(l-1, d-2)$[1]. As long as the set of all columns is bigger than the set of ineligible vectors at each stage l, we can continue the construction. This is guaranteed if $q^{n-k} > \mathrm{Vol}_q(n-1, d-2)$, which is equivalent to

$$q^k < \frac{q^n}{\mathrm{Vol}_q(n-1,d-2)} \tag{16}$$

as postulated. □

[1]inequality instead of equality as some linear combinations might be the same vector

Corollary 4. We have that

$$A_q(n,d) \geq \frac{q^n}{\mathrm{Vol}_q(n-1,d-2)} \tag{17}$$

Theorem 5 (Roth Theorem 4.5)**.** For the field $\mathbb{F}_q$ and for $n,k,d \in \mathbb{N}$, let ρ be given by

$$\rho = \frac{q^k-1}{q-1}.\frac{\mathrm{Vol}_q(n,d-1)}{q^n} \tag{18}$$

then all but a fraction of at most ρ of the linear (n,k) codes over $\mathbb{F}_q$ have minimum distance of at least d.

Proof. Given in *Introduction to Coding Theory* by Ron M. Roth, pp. 110. □

Corollary 6. For $q=2$, if

$$\mathrm{Vol}_q(n,d-1) \leq 2^{n-k-1}$$

then more than half of the linear (n,k) codes will have a minimum distance $\geq d$.

3 Asymptotic Bounds

In this section, we look into bounds that hold for the asymptotic case, i.e., $n \to \infty$.

Definition 3.1 (Normalized distance)**.** For any (n,k,d) code we define normalized distance (δ) as

$$\delta := \frac{d}{n} \tag{19}$$

♢

Definition 3.2 (Asymptotically good codes)**.** An *asymptotically good code* is a code family that has an asymptotically positive normalized distance

$$\lim_{n\to\infty} \delta = \lim_{n\to\infty} \frac{d}{n} > 0 \tag{20}$$

and a positive rate

$$\lim_{n\to\infty} R(\delta) = \lim_{n\to\infty} \frac{log_q A_q(n,\delta n)}{n} \geq 0 \tag{21}$$

♢

For the rest of the discussion in this lecture, we will take the case of $\mathbb{F}_2$ and find the relation between δ and $R(\delta)$.

Theorem 7 (Asymptotic singleton bound)**.** For any binary (n,d) code, the asymptotic rate $R(\delta)$ of the code is bounded by

$$\boxed{R(\delta) \leq 1-\delta} \tag{22}$$

Proof. We start with the original singleton bound that gives us

$$log_2 A_2(n, \delta n) \leq n - d + 1,$$

on dividing by n and applying the asymptotic condition, we get

$$\lim_{n \to \infty} \frac{log_2 A_2(n, \delta n)}{n} \leq \lim_{n \to \infty} \frac{n - d + 1}{n},$$

which corresponds to the equation

$$\mathbb{R}(\delta) \leq 1 - \delta.$$

□

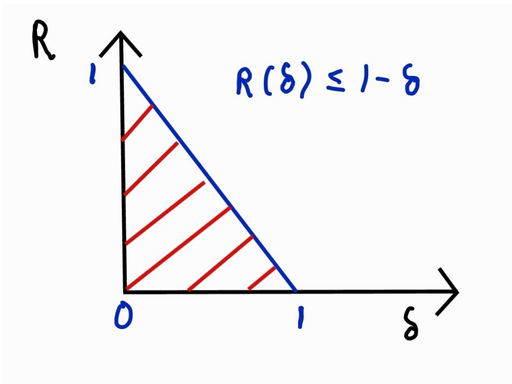

Figure 1: Rate vs δ for asymptotic singleton bound

Now, we look into the sphere packing (Hamming) bound for the asymptotic case. First, we start with the following definition-

Definition 3.3 (Binary entropy)**.** The Binary entropy $H(p)$ for any $p \in [0, 1]$ is defined as

$$H(p) = -p * log_2 p - (1 - p) * log_2(1 - p) \tag{23}$$

with $H(0) = H(1) = 0$. ♢

We know that volume of sphere $\text{Vol}_2(n, pn)$ is given by

$$\text{Vol}_2(n, pn) = \sum_{i=0}^{pn} \binom{n}{i}, \quad p < \frac{1}{2}. \tag{24}$$

Also, note that for the case of asymptotic conditions (check out *Roth* Lemma 4.7 and *Stirling approximation* for further details)-

$$\lim_{n \to \infty} \text{Vol}_2(n, pn) = \lim_{n \to \infty} \sum_{i=0}^{pn} \binom{n}{i} \approx 2^{H(p)} \tag{25}$$

Theorem 8 (Asymptotic singleton bound)**.** For any binary (n, d) code, the asymptotic rate $R(\delta)$ of the code is bounded by

$$\boxed{R(\delta) \leq 1 - H(\frac{\delta}{2})} \tag{26}$$

Proof. We start by looking toward the original Hamming bound expression that gave us

$$A_2(n, d) \leq \frac{2^n}{\mathrm{Vol}_2(n, \lceil \frac{d-1}{2} \rceil).} \tag{27}$$

For the asymptotic case of the above equation with $\lim_{n\to\infty}$ and $d = \delta n$, we take log on both sides and divide the entire equation by n to obtain

$$R(\delta) \leq \lim_{n\to\infty} 1 - \mathrm{Vol}_2(n, \lceil \frac{\delta n - 1}{2} \rceil), \tag{28}$$

which can be equivalently written as

$$R(\delta) \leq \lim_{n\to\infty} 1 - \frac{1}{n} log \binom{n}{\lceil \frac{\delta n - 1}{2} \rceil}, \tag{29}$$

Now, we apply the Stirling approximation to obtain the following expression for sphere packing for asymptotic conditions-

$$R(\delta) \leq 1 - H(\frac{\delta}{2}).$$

□

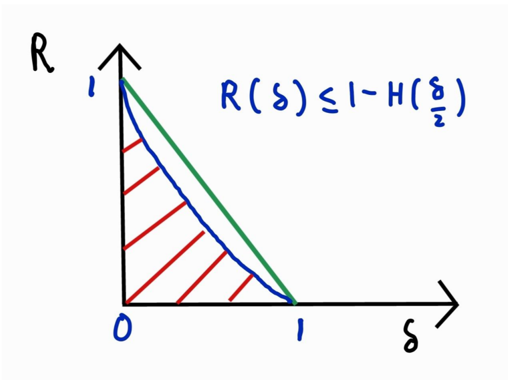

Figure 2: Rate vs δ for asymptotic Hamming bound